Buy flexibility where it helps.
Know when to withdraw.

Curtailment Action Lab — a decision workspace for grid planners and flexibility aggregators.

Same flexible demand. Different connection.
A recommendation with explicit conditions.

Real public network model · Hypothetical offers · Inspectable calculations

A cheap offer can be the wrong offer.

Price, network location, delivered flexibility and the operating point must be considered together.

1 / Compare Keep total demand and wind availability equal within each location comparison.

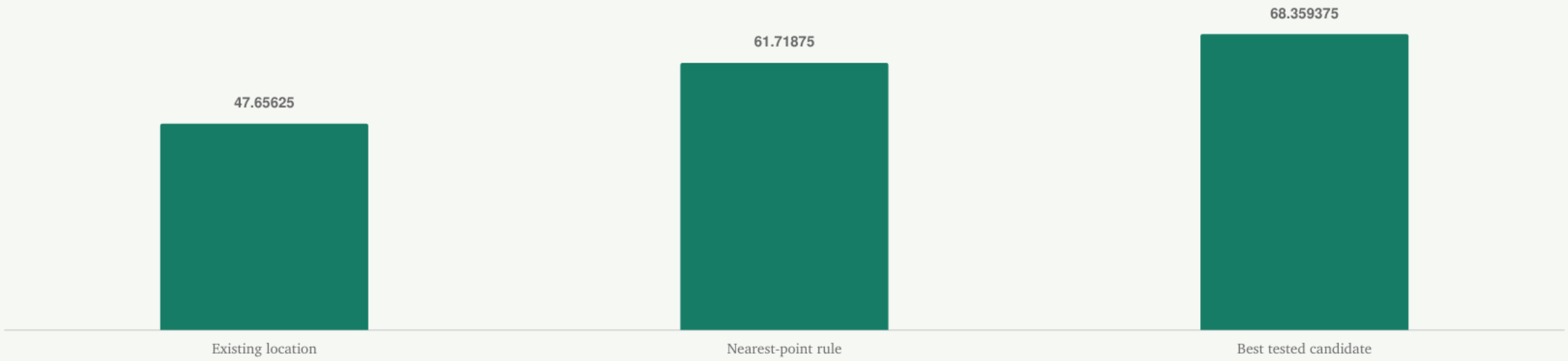
2 / Challenge Test reduced wind, partial delivery, changed demand and line state.

3 / Decide Retain, switch or withdraw; show the required data and calculation record.

MEASURED MODEL RESULT

The same 20 MW can enable more wind.

Best tested wind acceptance (MW) in the prepared Irish planning snapshot.



+20.70 MW vs existing location | +6.64 MW vs nearest-point rule

A headline result is not enough.

The nominal best candidate is tested again under changed conditions.

Half delivery	Added wind falls from 20.70 MW to 9.57 MW. This was recalculated, not divided by two.
Low available wind	At 50 MW available wind, added wind is 2.34 MW. Some assumed prices no longer pay.
+1% demand	The fixed-control baseline is unverified. The tool withdraws the recommendation if this test is required.

DEMONSTRATION

From a quote to a conditional decision.

Scenario Lab → Offer decisions

Set the offer

Enter assumed price, budget, energy envelope and minimum wind benefit.

Choose tests

Require the conditions that matter. Inspect each recorded pass or failure.

Review the result

See the chosen site, worst tested benefit, withdrawal reason and source records.

CONTROLLED EVALUATION

The checks change the recommendation.

288 authored decision cases; actual model records with synthetic prices and requirements.

74

Initial offers fail an extra required condition

7

Switched to a passing offer

67

Recommendations withdrawn

An enforced decision rule on a finite test set — not an AI accuracy or field-saving score.

Ask AI to challenge the recommendation.

A bounded tool interface; numerical evidence stays with the calculation engine.

The agent chooses	One allowed recorded condition at a time. It cannot change prices, limits or physical results.
The benchmark checks	Equal record-read budgets, first counterexample, rejected calls and provider time.
Connection status	Azure adapter and local contract tests are ready. Live AI contribution is not yet measured.

One evidence trail for several decisions.

A pilot begins with reviewed network limits and real, locatable flexibility offers.

Grid planner	Which candidate passes the reviewed network tests, and what input is still missing?
Aggregator / generator	Which assumed offer can meet a wind-gain requirement within budget?
Technology partner	An auditable planning component with a bounded AI interface and exportable records.